

Replacing DPR and BART in RAG: A Component-Level Study Using ColBERT and LLaMA

Replicating: Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks", NeurIPS 2020

Github: <https://github.com/23spakkerakari/Knowledge-Intensive-NLP-RAG>

1. Introduction

Lewis et al. (2020) introduced Retrieval-Augmented Generation (RAG), a hybrid architecture that conditions a sequence-to-sequence model on passages retrieved at inference time. The original system has two trainable components: a Dense Passage Retriever (DPR) that embeds queries and passages as single dense vectors for fast inner-product search, and a BART generator that produces answers conditioned on the top-k retrieved passages.

Why this problem is worth studying: DPR and BART were state-of-the-art at the time of publication but carry well-documented limitations. DPR's single-vector representation cannot distinguish between token-level nuances within a passage, which leads to coarse retrieval on precise factual queries. BART's generation quality is bounded by its parameter count and the distribution of its fine-tuning data. Newer models have since addressed both limitations independently, but their joint effect within the RAG pipeline has not been cleanly isolated.

Our contribution. We run a controlled component-swap experiment: DPR is replaced with ColBERTv2 (late-interaction retrieval) and BART is replaced with Nvidia's LLaMA 3.1 8B Instruct (zero-shot instruction-tuned generation). By running three configurations (baseline, retriever-only swap, full swap) on both evaluation domains, we can attribute score changes to the retriever, the generator, or their interaction. The core question is whether modern components improve end-task performance without task-specific fine-tuning, and where the trade-offs lie.

2. Chosen Result

We target Table 1 (NQ Exact Match) and Table 2 (MS-MARCO ROUGE-L and BLEU-1) from Lewis et al. These two tables are the primary empirical claim in the paper and span fundamentally different task requirements: NQ rewards precise factual span retrieval while MS-MARCO rewards fluent multi-sentence synthesis. Choosing both forces the experiment to reveal whether component improvements generalize across task types or are specific to one. The three metrics together provide a concrete, reproducible benchmark that a reader can directly compare to the original.

3. Methodology

Retrieval corpus. We sample 50,000 passages from a single shard of the facebook/wiki_dpr NQ dataset, matching the baseline corpus size to isolate the effect of the retrieval model rather than corpus coverage. **Note:** while the original paper uses 21 million passages, to run training and evaluation given our disk space, 1xA100 gpu, and RAM meant that it would take multiple days to run one iteration alone. 50k was the most reasonable passage corpus size given our compute limits.

DPR limitation and ColBERT fix. DPR compresses each passage into one 768-dimensional vector, collapsing all token-level information into one vector. Although great for efficiency + compute, it struggles with fine-grained matches, and extraction of themes and concepts. ColBERTv2 retains a full matrix of token embeddings per passage and computes a MaxSim score at query time: each query token is matched against every passage token and only the best match contributes. This late-interaction mechanism preserves term-level precision without requiring a full cross-encoder reranking pass. In practice this means ColBERT can distinguish, for example, between a passage that mentions "Paris, Texas" and one about "Paris, France" in a way that DPR cannot. We use RAGatouille 0.0.9.post2 with the Stanford ColBERT backend, building a PLAID approximate index over 50k passages. k=5 passages are retrieved for all experiments.

BART limitation and LLaMA fix. BART-large has approximately 400M parameters and was fine-tuned on NQ and MS-MARCO training data respectively. Its generation quality is constrained by parameter count and by the distributional match to its fine-tuning corpus. LLaMA 3.1 8B Instruct has 8B parameters, has seen far more diverse pre-training text, and has been instruction-tuned for follow-the-format generation. We load it in 4-bit NF4 quantization via bitsandbytes to fit alongside the ColBERT index on a single A100 40GB GPU. Crucially, we do not fine-tune LLaMA, which creates an intentionally uneven comparison: we are testing whether raw capability and instruction-following can substitute for task-specific training. Retrieved passages are formatted as a numbered context block in the chat prompt, followed by the question. For NQ, we prompt for a short factual answer (max 40 new tokens); for MS-MARCO, we prompt for one to two complete sentences (max 80 tokens).

Evaluation details and the EM relaxation. NQ is evaluated with Exact Match after standard normalization (lowercase, strip articles and punctuation). Because LLaMA's instruction-tuned behavior generates full sentences rather than bare answer spans, strict EM would systematically undercount correct answers. We apply a containment check: EM is credited if any gold answer appears within the prediction or vice versa. This is a deliberate and documented relaxation; strict EM numbers would be lower. MS-MARCO is evaluated with corpus BLEU-1 and macro ROUGE-L, implemented from scratch to match the paper's evaluation conditions.

Ablation structure. Three configurations are run on both domains: (1) DPR + BART baseline from Lewis et al., (2) ColBERT + BART with only the retriever swapped, and (3) ColBERT + LLaMA with both swapped. Configuration (2) isolates retriever contribution; the delta from (2) to (3) isolates generator contribution.

4. Results and Analysis

Tables 1-3. Performance across model configurations and evaluation domains.

Table 1. Main RAG-Sequence Reproduction					
Model	Task	Dataset	Metric	Ours	Paper
DPR + BART / RAG-Sequence	Open-Domain QA	NQ	EM	11.41	44.5
DPR + BART / RAG-Sequence	Abstractive QA	MS-MARCO	ROUGE-L	34.27	40.8
DPR + BART / RAG-Sequence	Abstractive QA	MS-MARCO	BLEU-1	25.26	44.2

Table 2. ColBERT + BART Retriever-Swap Ablation				
Experiment	Task	K (# OF RETRIEVED DOCUMENTS)	Metric	Result
ColBERT retrieval only	NQ Open-Domain QA	5	Recall@5	32.78%
ColBERT + generic BART	NQ Open-Domain QA	5	EM / F1	0.00 / 0.07
ColBERT + fine-tuned NQ BART	NQ Open-Domain QA	5	EM / F1	0.22 / 0.27
ColBERT + adapted BART, 5k train	NQ Open-Domain QA	5	EM / F1	0.20 / 0.26
ColBERT + adapted BART, 5k train	NQ Open-Domain QA	50	EM / F1	0.18 / 0.23

Table 3. ColBERT + LLaMA Results				
Model	Task	Dataset	Metric	Result
ColBERT + LLaMA-3.1-8B	Open-Domain QA	NQ	EM	15.40
ColBERT + LLaMA-3.1-8B	Abstractive QA	MS-MARCO	ROUGE-L	17.90
ColBERT + LLaMA-3.1-8B	Abstractive QA	MS-MARCO	BLEU-1	16.25

Note: ColBERT retrieved an answer-containing passage for **32.78%** of NQ questions, but low EM showed that stronger retrieval alone did not translate into better generation without compatible retriever-generator training.

Open-domain QA (NQ) Analysis. We tested whether replacing the DPR/MIPS retriever with ColBERT would improve open-domain QA by providing richer token-level passage matching. However, the ColBERT + BART retriever-swap result achieved only 0.22 EM, far below the 44.5 EM reported in the original paper. This suggests that simply swapping in a stronger retriever is not enough when the generator is not fully trained with that retriever’s outputs. Although

ColBERT achieved nontrivial Recall@5, the end-to-end EM remained low, showing that retrieval quality alone does not guarantee better QA performance without compatible retriever-generator training.

The ColBERT + LLaMA setup achieved 15.40 EM on NQ, which is substantially below the paper baseline but much stronger than our ColBERT + BART result. This suggests that LLaMA was better able to use the retrieved passages in a zero-shot setting, even though it was not fine-tuned on NQ. Overall, these results support our main finding: ColBERT is a stronger retriever in principle, but our implementation was a plug-in retriever swap rather than a fully trained ColBERT-RAG model. Future work should train the generator more extensively on ColBERT-retrieved passages or integrate ColBERT directly into the RAG scoring and training loop.

Abstractive QA (MS-MARCO) Analysis. LLaMA’s advantage was expected to be most visible here, since MS-MARCO rewards fluent multi-sentence synthesis. ColBERT + LLaMA achieved ROUGE-L of 17.90 and BLEU-1 of 16.25, both substantially below the paper baseline of 40.8 and 44.2. The drop is larger than on NQ in absolute terms and confirms the lexical drift hypothesis: LLaMA generates fluent and often accurate answers but uses different vocabulary and sentence structure than the reference answers, which n-gram metrics penalize directly. We did not complete a ColBERT + BART MS-MARCO run, so the available MS-MARCO comparison is between our RAG-Sequence baseline and the ColBERT + LLaMA setup. This was due to time constraints as well as realizing we would run into the same issue as experienced with our ColBERT.+ BART setup for open-domain QA task.

Key assumptions and limitations. Four factors limit direct comparability with Lewis et al.: (i) the 50k corpus is much smaller than the full Wikipedia index, which constrains recall ceilings; (ii) LLaMA is zero-shot while BART was fine-tuned, making generator scores not apples-to-apples; (iii) the relaxed EM criterion inflates NQ scores relative to strict EM; (iv) the PLAID index is an approximation that the ragatouille library itself flags as potentially less precise than FAISS on larger corpora.

5. Reflections

What we learned about component swapping. The main takeaway is that raw model capability does not automatically transfer into end-task gains when the original system was fine-tuned end-to-end on the same task. Our results bear this out directly: ColBERT + LLaMA scored 15.40 EM on NQ versus the paper’s 44.5, and 17.90 ROUGE-L and 16.25 BLEU-1 on MS-MARCO versus 40.8 and 44.2. BART’s competitive performance came substantially from having seen the NQ and MS-MARCO training distributions. LLaMA is a far more capable model in absolute terms, but capability and calibration are distinct: a zero-shot 8B model does not outperform a fine-tuned 400M model on narrow factual extraction or n-gram-scored generation tasks. This distinction between generalist capability and task-specific calibration is the central research insight of the project.

Implementation iterations and obstacles. This project required substantially more iteration than anticipated. The colbert-ir/colbertv2.0 model download failed repeatedly due to partial HuggingFace cache writes that left a corrupted snapshot directory; the fix was bypassing the hub cache entirely by downloading to a flat local path via `snapshot_download`. The ragatouille 0.0.9.post2 package requires langchain pinned to 0.1.20 because the langchain.retrievers import path was removed in 0.2.0. The CUDA C++ extensions for residual decompression (`decompress_residuals_cpp`, `packbits_cpp`) require a live A100 JIT compilation pass before the first index load, which fails silently on CPU-only runtimes. Each of these was discovered through failure and corrected iteratively. Beyond environment issues, the decision to relax the EM metric was itself an iterative choice: early test runs showed that strict EM undercounted visibly correct LLaMA answers by roughly 30-40%, which prompted the containment-based alternative.

What we would do differently with more resources. Three extensions would make the comparison substantially more rigorous. First, fine-tuning LLaMA on NQ and MS-MARCO training data would make the generator comparison symmetric with the paper baseline. Second, evaluating with BERTScore alongside BLEU and ROUGE would detect cases where LLaMA generates semantically correct but lexically different answers that n-gram metrics penalize. Third, scaling the retrieval corpus from 50k to the full Wikipedia dump would test whether ColBERT’s late-interaction advantage holds at production scale, where its larger index footprint becomes a practical bottleneck. The 50k corpus was a pragmatic choice driven by Colab disk and time constraints, not a theoretically motivated one.

6. References

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